

Terhal • ترحال

Swipe a region into a trip, then book the person who knows it.

A bilingual travel app for Ma'an, Jordan — and a second app for the guides who work there.

Role Full-stack · data pipeline, backend, app **Team** 3 **2026**

What it is

A mobile app for visitors to Ma'an, the Jordanian governorate that contains Petra. Nearly all of the region's tourism revenue lands at one monument while the rest of the governorate stays invisible, and Ma'an carries the highest unemployment rate in Jordan at 29.4%. Terhal is an attempt at the connective tissue: help a visitor discover more than the one famous site, turn that into a real plan, and put them in front of a verified local guide at a price neither side has to argue about.

How it works

Planning is a card deck rather than a form. You swipe through the region's destinations and the app builds a day-by-day timeline from the ones you kept — ordered by geography, packed against a clock from 08:00 to 18:00, with real driving times between stops and anything that will not fit handed back with a reason instead of silently dropped. The scheduler is plain arithmetic rather than a model call, so the same plan comes out every time.

Booking is the second half. Verified guides appear nearest-first on a map, each quoting what *this* trip would cost. A guide is 25 JOD an hour — one rate, the same for everyone, no call-out fee and no group surcharge — which is the whole fairness argument in a single number. What is shown before booking is an estimate; the trip is metered between the guide starting and ending it, so one that finishes early costs less. Starting it needs the visitor's PIN, typed by the guide in front of them, which is what proves the meeting actually happened.

The data underneath

48 places, 23 of them top-level destinations, each with bilingual descriptions, coordinates, researched history and honest accessibility notes, illustrated with 42 CC-licensed photographs carrying their attribution. Nothing is invented: places that could not be sourced are flagged and left out, and the pipeline reports what it skipped every time it runs.

That discipline caught real errors. The accessibility filter originally matched the phrase “not accessible” and so excluded Petra entirely — even though the research says its main trail is wheelchair-passable and only the climbs beyond are not. Inverting it to default-deny, where a place counts as accessible only when the research affirmatively says so, leaves four places out of twenty-three. That is a worse-looking number and the true one.

Built with

- React, Vite, Tailwind — one build, two apps
- FastAPI, SQLAlchemy, Alembic
- PostgreSQL on Supabase
- Leaflet and OpenStreetMap
- Multimodal vision and chat, grounded on the data

Scale

- ~7,800 lines of Python, ~8,000 of JSX
- 48 places, 42 licensed images, 12 advisors
- 33 backend tests over pricing and scheduling
- Arabic and English throughout, RTL included
- Built for the Ma'an Hackathon (Irada Program)

The app



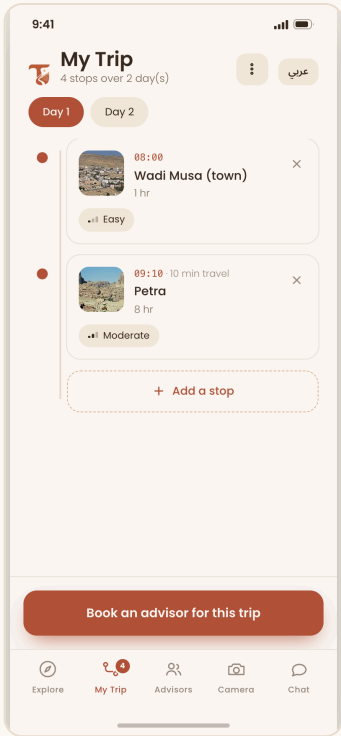
Welcome

Opens in English or Arabic; the whole interface mirrors for RTL.



Swipe to plan

Swipe right to add a place, left to skip. Cards carry the real visit duration, difficulty, and whether a guide is advised.



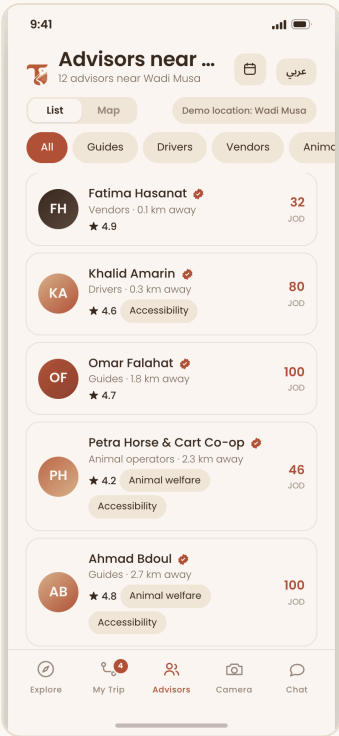
The generated plan

A day-by-day timeline built from what was swiped: stops ordered by geography, with real drive times between them.



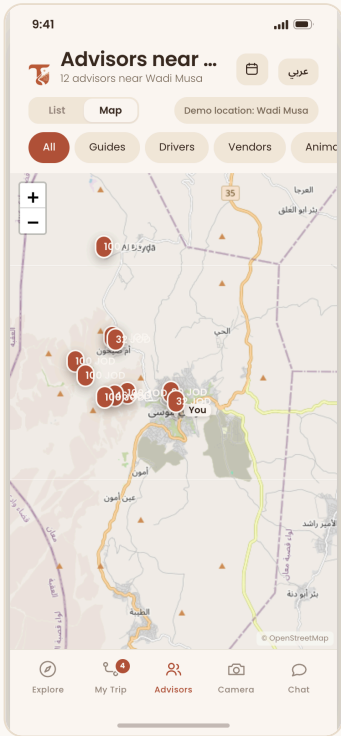
The same screen in Arabic

Not a translation layer bolted on at the end — layout, navigation and the timeline rail all mirror.



Verified local advisors

Guides, drivers and vendors near the visitor, each showing what this trip costs before anything is committed to.

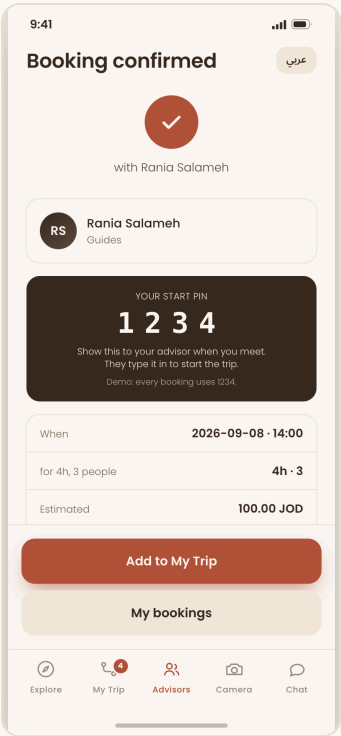


Advisors on the map

Distance-sorted, price on the pin. The alternative to negotiating at the gate with whoever approaches first.

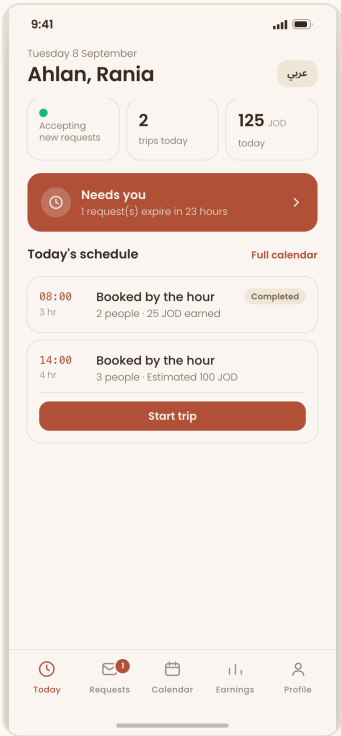
Screenshots of the running application. Prices are labelled in the product as demo figures; advisor accounts are seeded, and every photograph carries its licence and attribution.

The app, continued



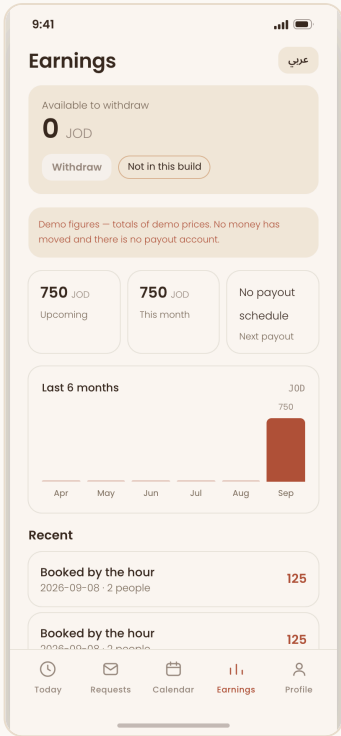
Booking, with the meeting PIN

The visitor holds a PIN. The guide has to be told it in person and type it in, which is what proves the two actually met.



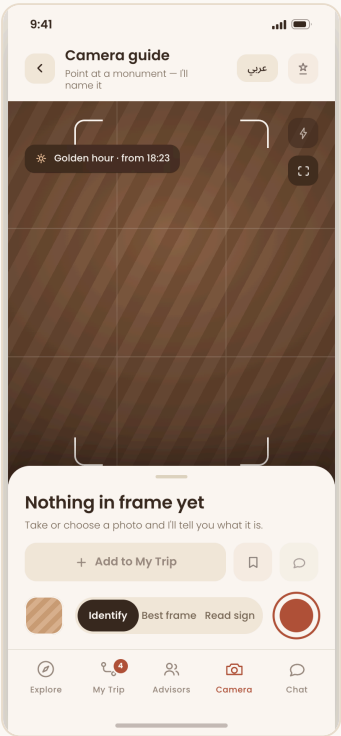
The guide's day

A second app on the same backend. One trip finished and settled at what it actually ran to; the next waiting on the PIN.



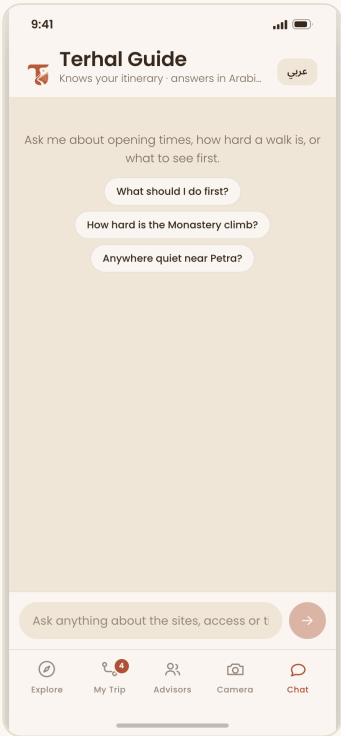
Guide earnings

Every figure derived from real bookings, and labelled as demo pricing rather than dressed up as revenue.



Camera guide

Point the phone at a monument and it identifies it, then narrates from a researched script rather than improvising.



Bilingual assistant

Answers grounded in the same researched dataset, so it declines to invent what the research does not cover.

Screenshots of the running application. Prices are labelled in the product as demo figures; advisor accounts are seeded, and every photograph carries its licence and attribution.